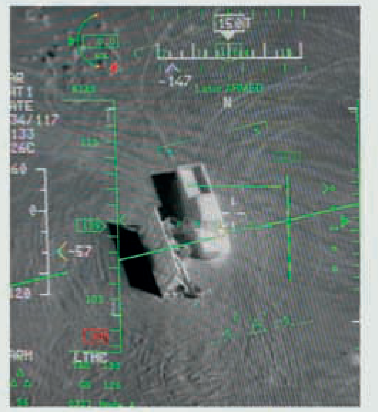
> “Our **moral authority** is as **important** [as] our **troop strength** and our **high-tech weapons**”
>
> **ROBERT REICH, FORMER US SECRETARY OF LABOR**, JULY 2004

Laser weapons have evolved into two distinct categories. Laser rifles, such as the American TR3 and PHASR, cause temporary blindness, and are intended to discourage rather than kill. A second line of development consists of very powerful lasers, such as the experimental Laser Avenger, which is designed to shoot down incoming missiles.

For the soldier on the battlefield, the conflict in Afghanistan has been a testing ground for a whole range of new weapons. The XM-26 LSS under-barrel shotgun – which attaches to an M4 or M16-type assault rifle – enables door-breaching and close-range firepower, and can fire non-lethal projectiles such as tear gas. Another innovation is the H&K XM25 Counter-defilade Target Engagement (CDTE). This 25mm semi-automatic grenade launcher uses a “smart” laser range finder to ensure precise detonation of the grenade to a range of up to 1,000m (1,090 yards). The grenade measures the distance it travels by counting its own rotations.



▲ AN MQ-9 REAPER IMAGE
A display screen in a ground control station at a US Air Force base shows the view captured by a MQ-9 Reaper camera during a training mission.

◄ A TALON 3B ROBOT
A claymore land mine is removed from a sand dune by a Talon 3B robot during a training exercise in Bahrain. Designed to search and destroy Improvised Explosive Devices (IEDs), the robot is operated remotely by technicians using monitors and video equipment attached to the unit.

KEY **EVENTS**

1970–PRESENT

- **1972** The Wheelbarrow bomb-disposal robot is developed for the British Army for use in Northern Ireland.
- **1982** The Lockheed F-117, the first aircraft to feature stealth technology, is adopted by the US Air Force (see pp.412–13).
- **1994** The Predator UAV makes its first flight. It is designed for reconnaissance only.
- **2001** The Reaper UAV is developed. It is larger and more versatile than the Predator.
- **2006** The F-35 Lightning II – a jet fighter with advanced stealth technology – goes into production.
- **2007** Reaper UAVs begin combat missions over Afghanistan.
- **2009** Combat field trials of the IAWS take place in Iraq and Afghanistan.
- **2010** Metaflex “invisible” material is developed.